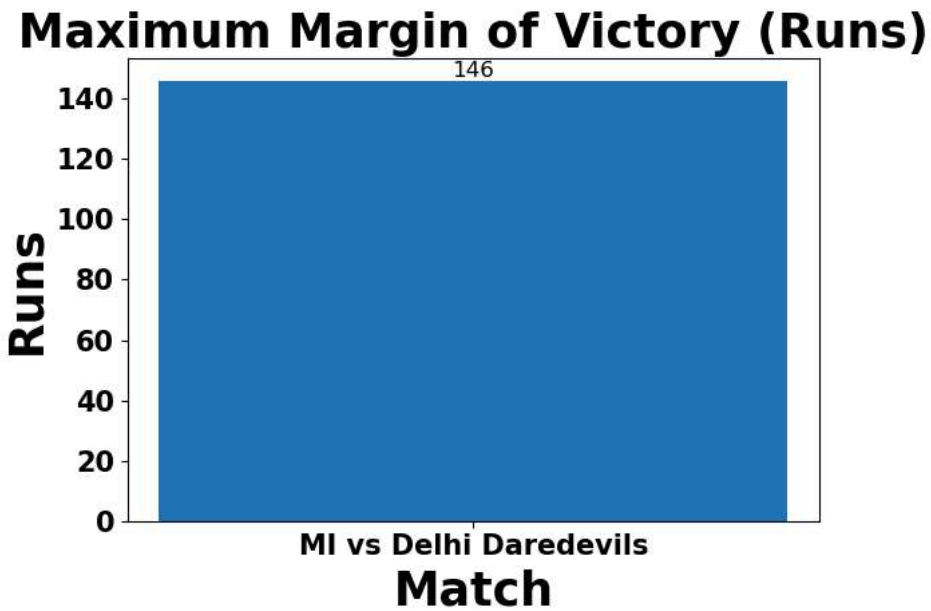
1. Matches won by maximum margin of runs.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("matches.csv")

max_margin = df['win_by_runs'].max()

max_match = df[df['win_by_runs'] == max_margin].iloc[0]

winner = max_match['winner']
team1 = max_match['team1']
team2 = max_match['team2']

opponent = team2 if winner == team1 else team1

label = f"{winner} vs {opponent}"

plt.figure()
bars = plt.bar(label, max_margin)

plt.xlabel("Match",fontweight='bold',fontsize=25)
plt.ylabel("Runs",fontweight='bold',fontsize=25)
plt.title("Maximum Margin of Victory (Runs)",fontweight='bold',fontsize=25)

plt.text(label, max_margin, str(max_margin),
         ha='center', va='bottom', fontsize=12)

plt.xticks(rotation=0,fontweight='bold',fontsize=15)
plt.yticks(rotation=0,fontweight='bold',fontsize=15)
plt.tight_layout()

plt.show()
```



Insights

- The highest victory margin is 146 runs.
- It indicates a highly one-sided match
- The winning team (Mumbai Indians) completely dominated the opponent (Delhi Daredevils).
- This is an exceptional performance compared to other matches.

2. Match won by maximum wickets.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("matches.csv")

max_wickets = df['win_by_wickets'].max()

max_match = df[df['win_by_wickets'] == max_wickets].iloc[0]

winner = max_match['winner']
team1 = max_match['team1']
team2 = max_match['team2']

opponent = team2 if winner == team1 else team1

label = f"{winner} vs {opponent}"

plt.figure()
plt.bar(label, max_wickets)

plt.xlabel("Match",fontweight='bold',fontsize=15)
plt.ylabel("Wickets",fontweight='bold',fontsize=15)
plt.title("Maximum Wicket Margin Victory",fontweight='bold',fontsize=15)

plt.text(label, max_wickets, str(max_wickets),
         ha='center', va='bottom', fontsize=12)

plt.xticks(rotation=0,fontsize=12)
plt.yticks(rotation=0,fontsize=12)
plt.tight_layout()

plt.show()
```

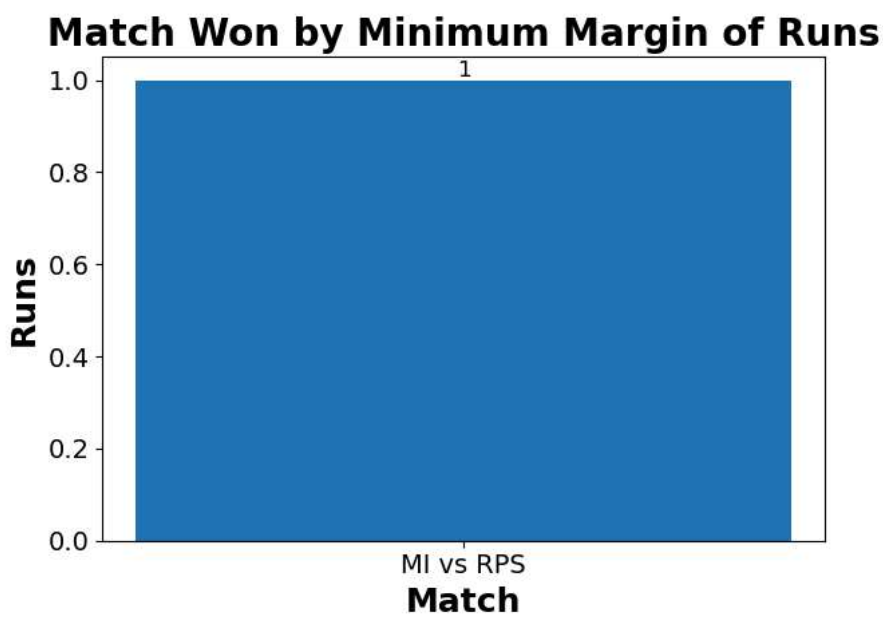
Insights

- The maximum wicket margin recorded is 10 wickets.
- The match was won by Kolkata Knight Riders against Gujarat Lions.
- This type of win is considered very rare and highly one-sided in T20 cricket.
- The winning team chased the target without losing any wicket.
- It shows strong batting and bowling performance by Kolkata Knight Riders.

3.Match won by the minimum margin of runs.

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("matches.csv")

runs_wins = df[df['win_by_runs'] > 0]

min_margin = runs_wins['win_by_runs'].min()

min_match = runs_wins[runs_wins['win_by_runs'] == min_margin].iloc[0]

winner = min_match['winner']
team1 = min_match['team1']
team2 = min_match['team2']

opponent = team2 if winner == team1 else team1

label = f"{winner} vs {opponent}"

plt.figure()
plt.bar(label, min_margin)

plt.xlabel("Match",fontweight='bold',fontsize=18)
plt.ylabel("Runs",fontweight='bold',fontsize=18)
plt.title("Match Won by Minimum Margin of Runs",fontweight='bold',fontsize=20)

plt.text(label, min_margin, str(min_margin),
         ha='center', va='bottom', fontsize=12)

plt.xticks(rotation=0,fontsize=14)
plt.yticks(rotation=0,fontsize=14)
plt.tight_layout()
plt.show()
```



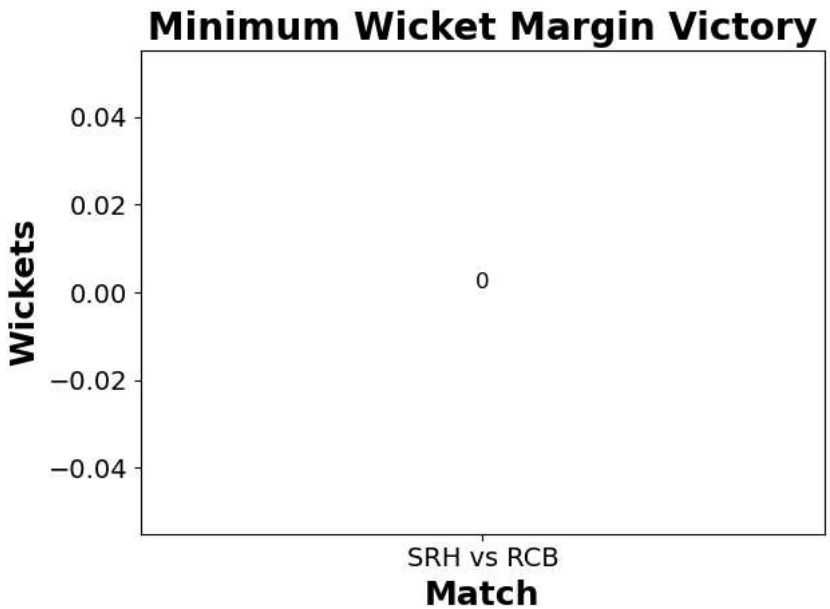
Insights

- This match shows Mumbai Indians beats Rising Pune Super Giants by one runs.
- This match was highly dominated by both the teams.
- A 1-run victory indicates a high-pressure and thrilling finish.
- Such wins highlight the importance of every single run in T20 cricket.

4.Match won by minimum wickets.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("matches.csv")

min_wickets = df['win_by_wickets'].min()
```

```
min_match = df[df['win_by_wickets'] == min_wickets].iloc[0]

winner = min_match['winner']
team1 = min_match['team1']
team2 = min_match['team2']

opponent = team2 if winner == team1 else team1

label = f"{winner} vs {opponent}"

plt.figure()
plt.bar(label, min_wickets)

plt.xlabel("Match",fontweight='bold',fontsize=18)
plt.ylabel("Wickets",fontweight='bold',fontsize=18)
plt.title("Minimum Wicket Margin Victory",fontweight='bold',fontsize=20)

plt.text(label, min_wickets, str(min_wickets),
         ha='center', va='bottom', fontsize=12)

plt.xticks(rotation=0,fontsize=14)
plt.yticks(rotation=0,fontsize=14)
plt.tight_layout()

plt.show()
```



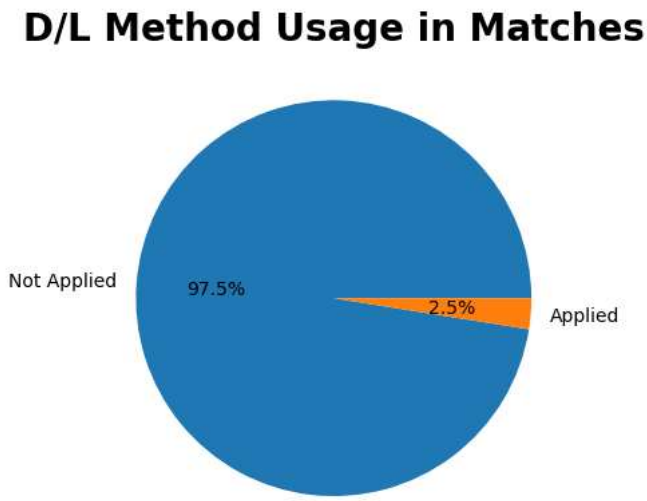
Insights

- In this match SRH beats RCB by 0 wickets.
- The game was likely decided in the final overs.
- Such victories are rare and highly competitive.
- The graph highlights how intense and tightly contested the match was.

5.Matches where D/L method was and wasn't applied.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("matches.csv")

dl_counts = df['dl_applied'].value_counts()

labels = ['Not Applied', 'Applied']

plt.figure()
plt.pie(dl_counts.sort_index(), labels=labels, autopct='%1.1f%%')
```

```
plt.title("D/L Method Usage in Matches",fontweight='bold',fontsize=20)

plt.show()
```

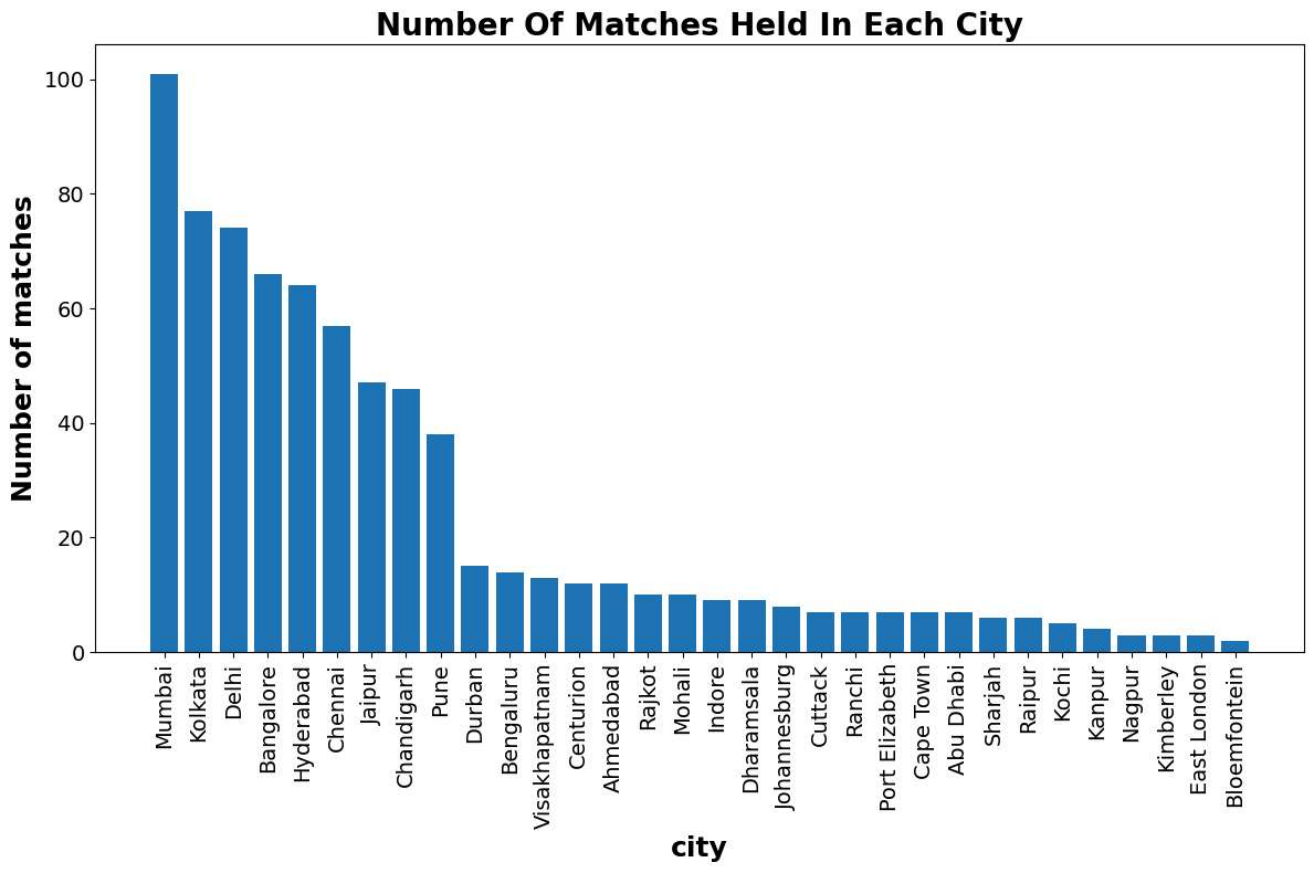


Insights

- This graph shows that in 97.5% matches the DLS method was not applied which means rain intrruption is very rare in given dataset.
- Most of the matches are completed in normal weather.
- This graph shows the venue selection is good and it avoid heavy rains.
- Result of most matches are not affected by D/L method

6.No. of matches held in each city.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("matches.csv")

city_counts = df['city'].value_counts()

plt.figure(figsize=(12,8))
plt.bar(city_counts.index, city_counts.values)

plt.xlabel("city",fontweight='bold',fontsize=18)
plt.ylabel("Number of matches",fontweight='bold',fontsize=18)
plt.title("Number Of Matches Held In Each City",fontweight='bold',fontsize=20)

plt.xticks(rotation=90,fontsize=14)
plt.yticks(rotation=0,fontsize=14)
plt.tight_layout()
plt.show()
```

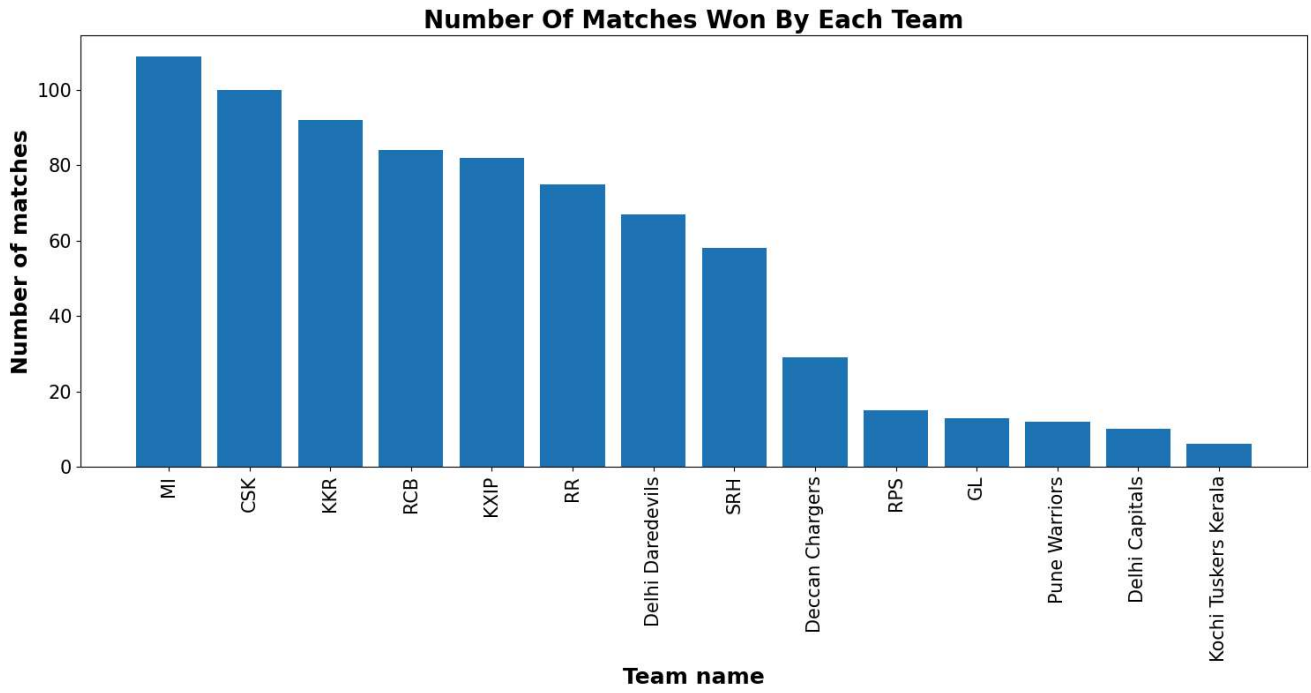


Insights

- Mumbai hosts most numbers of matches as compared to other cities. Mumbai hosts 100 matches.
- This graph shows some of the matches are played outside the country.
- Big metro cities like Banglore, Delhi, Chennai, Mumbai, Pune, etc hosts the most matches which helps to boost their economy.
- Their is sudden decline in number of matches after 6-8 big cities which shows uneven distribution of venues.

7.No. of matches won by each team.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("matches.csv")

winner_counts = df['winner'].value_counts()

plt.figure(figsize=(15,8))
plt.bar(winner_counts.index, winner_counts.values)

plt.xlabel("Team name",fontweight='bold',fontsize=18)
plt.ylabel("Number of matches",fontweight='bold',fontsize=18)
plt.title("Number Of Matches Won By Each Team",fontweight='bold',fontsize=20)

plt.xticks(rotation=90,fontsize=15)
plt.yticks(fontsize=15)
plt.tight_layout()
plt.show()
```

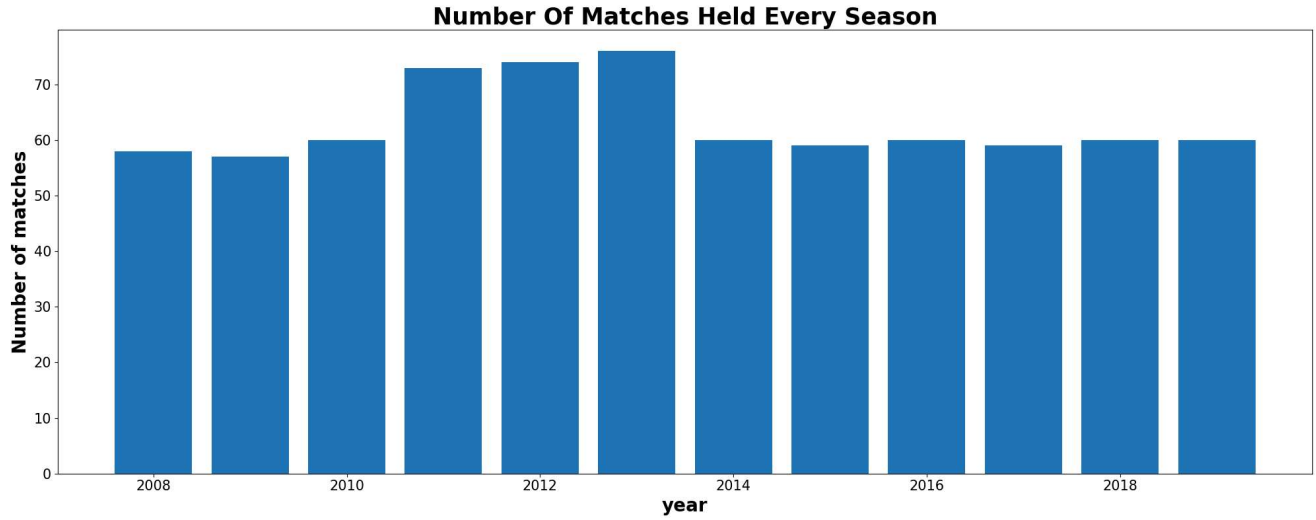


Insights

- Mumbai Indias has won most matches more than 100 matches in overall tournament.
- This graph shows MI, CSK, KKR, RCB, KxiP and RR highly dominates in the tournament.
- Last 6-7 teams are seen as weak teams in the tournament.
- The management team and coaches of first 7-8 teams are experienced as compared to last 6-7 teams.

8.No. of matches held every season.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("matches.csv")

season_counts = df['season'].value_counts()

plt.figure(figsize=(20,8))
plt.bar(season_counts.index, season_counts.values)

plt.xlabel("year", fontsize=20,fontweight='bold')
plt.ylabel("Number of matches",fontsize=20,fontweight='bold')
plt.title("Number Of Matches Held Every Season",fontsize=25,fontweight='bold')

plt.xticks(rotation=0,fontsize=15)
plt.yticks(fontsize=15)
plt.tight_layout()
plt.show()
```

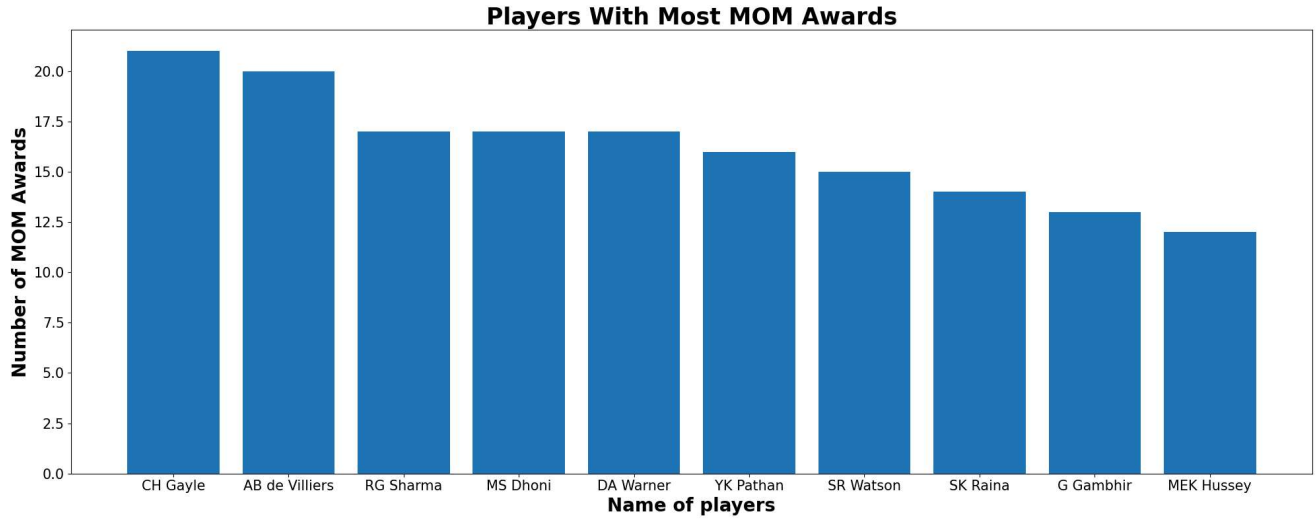


Insights

- From 2008 to around 2012–2013, the number of matches steadily increased. It shows the popularity of tournament continiously increasing.
- The graph of year 2011, 2012, 2013 was high which shows number of matches played in this seasons are more as compared to other seasons.
- Highest number of matches was recorded in year 2013 around 75 matches.
- After year 2013, the number of matches slightly decreased and stabilized around 60 matches per season.

9.Top 10 players based on no. of Man of Match (MOM) awards won.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("matches.csv")

top_players = df['player_of_match'].value_counts().head(10)

plt.figure(figsize=(20,8))
plt.bar(top_players.index, top_players.values)

plt.xlabel("Name of players",fontsize=20,fontweight='bold')
plt.ylabel("Number of MOM Awards",fontsize=20,fontweight='bold')
plt.title("Players With Most MOM Awards",fontsize=25,fontweight='bold')

plt.xticks(rotation=0,fontsize=15)
plt.yticks(rotation=0,fontsize=15)
plt.tight_layout()
plt.show()
```

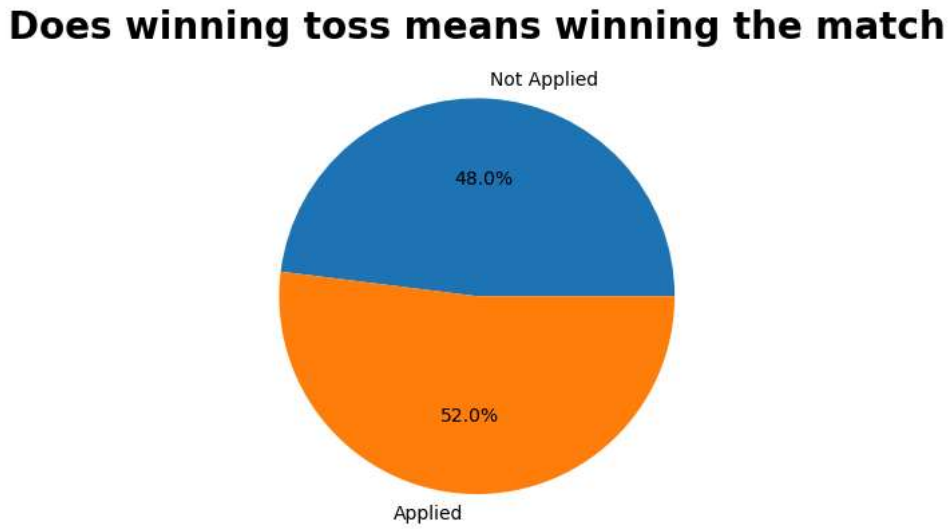


Insights

- This graph shows that these players has highly dominate for there respective teams.
- Chris gayle has won most man of the match awards it shows his performance and domination in this tournament.
- The graph shows that MS Dhoni, Rohit Sharma and David Warner has won equal man of the match award.
- According to graph there is very less margin in number of awards. It shows there is tough competetion in this area.

10.Does winning the toss mean winning the match?

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("matches.csv")

df['toss_win_match_win'] = df['toss_winner'] == df['winner']

result_counts = df['toss_win_match_win'].value_counts().sort_index()

lables = ['lost after winning toss', 'won after winning toss']

plt.figure()
plt.pie(result_counts, labels=labels, autopct='%1.1f%%')

plt.title("Does winning toss means winning the match",fontweight='bold',fontsize=20)

plt.show()
```

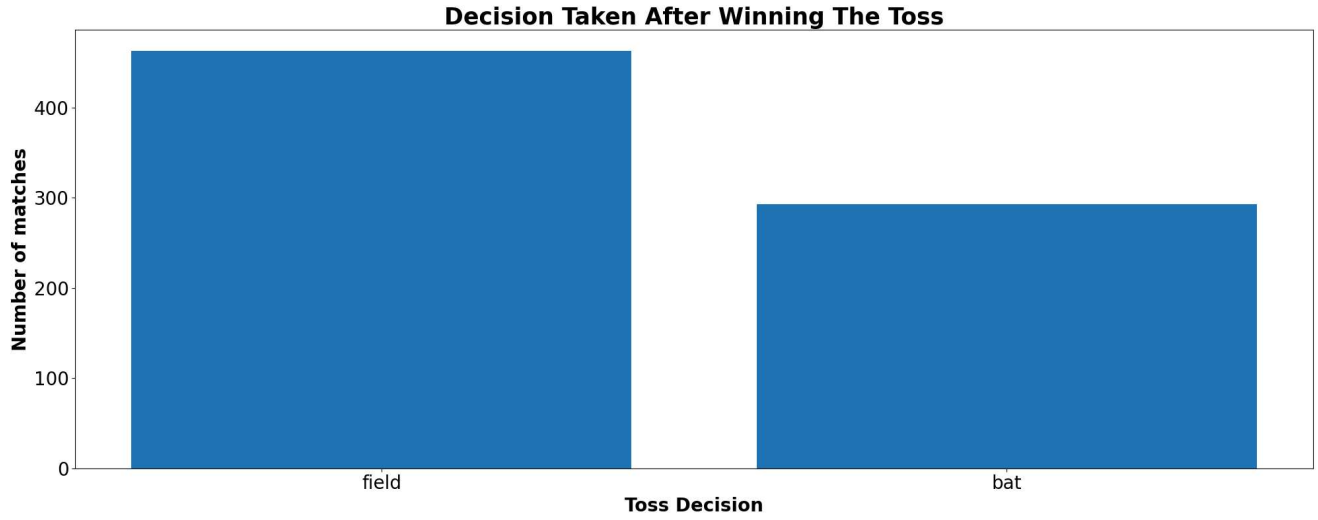


Insights

- This graph shows winning the toss doesn't means winning the match.
- This graph shows only 52% matches are won after winning the toss.
- According to this this graph the result of match was not depended on the result of toss.
- 48% of the matches are won after losing the toss. It means you can win the match after losing the toss.

11.What was the decision taken by captains when they won the toss?

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("matches.csv")

decision_counts = df['toss_decision'].value_counts()

plt.figure(figsize=(20,8))
plt.bar(decision_counts.index, decision_counts.values)

plt.xlabel("Toss Decision",fontsize=20,fontweight='bold')
plt.ylabel("Number of matches",fontsize=20,fontweight='bold')
plt.title("Decision Taken After Winning The Toss",fontsize=25,fontweight='bold')

plt.xticks(rotation=0,fontsize=20)
plt.yticks(rotation=0,fontsize=20)
plt.tight_layout()
plt.show()
```

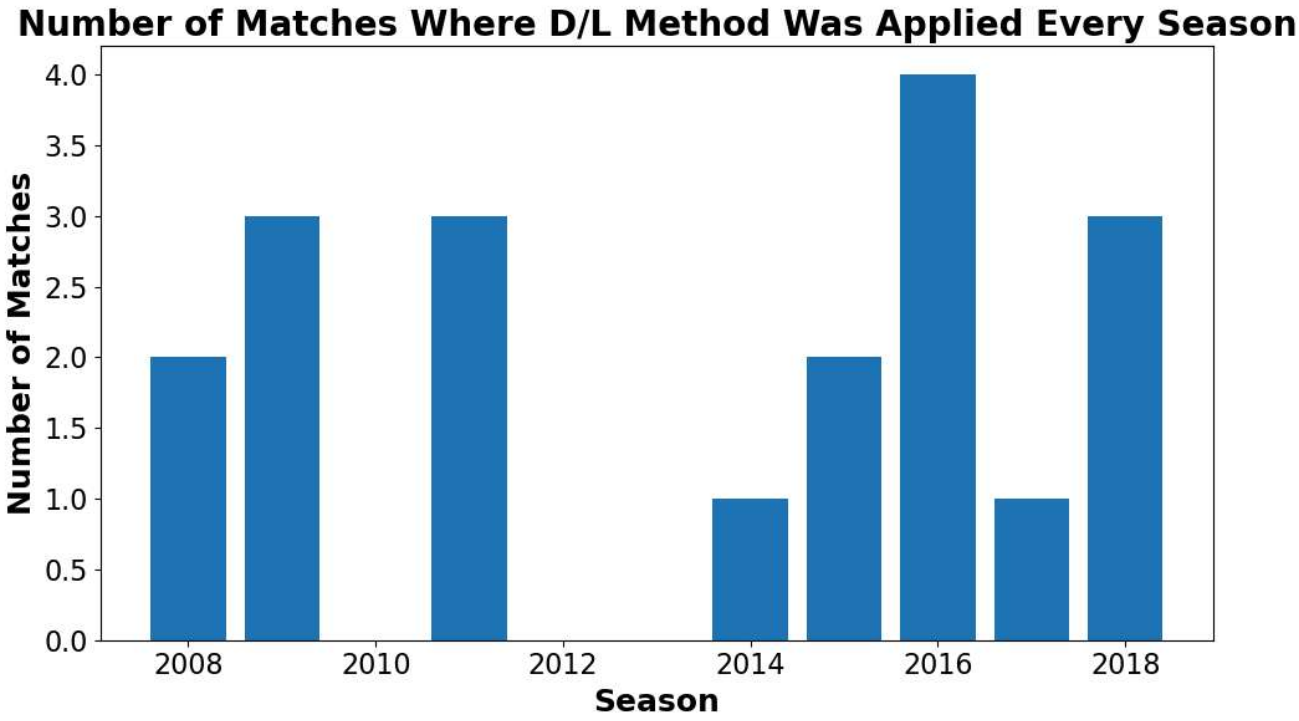


Insights

- This graph shows most of the captains choose field after winning the toss.
- Choosing field after winning toss shows most of the teams wants to chase the score.
- More than 400 times captains choose to field and around 300 times they choose to bat.
- According to this graph it seens that chasing the target is easy for winning the match.

12.No. of matches where D/L method was applied every season.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("matches.csv")

dl_matches = df[df['dl_applied'] == 1]

dl_per_season = dl_matches.groupby('season').size()

plt.figure(figsize=(10,6))
plt.bar(dl_per_season.index, dl_per_season.values)

plt.xlabel("Season",fontweight='bold',fontsize=18)
plt.ylabel("Number of Matches",fontweight='bold',fontsize=18)
plt.title("Number of Matches Where D/L Method Was Applied Every Season",fontweight='bold',fontsize=20)

plt.xticks(rotation=0,fontsize=16)
plt.yticks(rotation=0,fontsize=16)
plt.tight_layout()
plt.show()
```

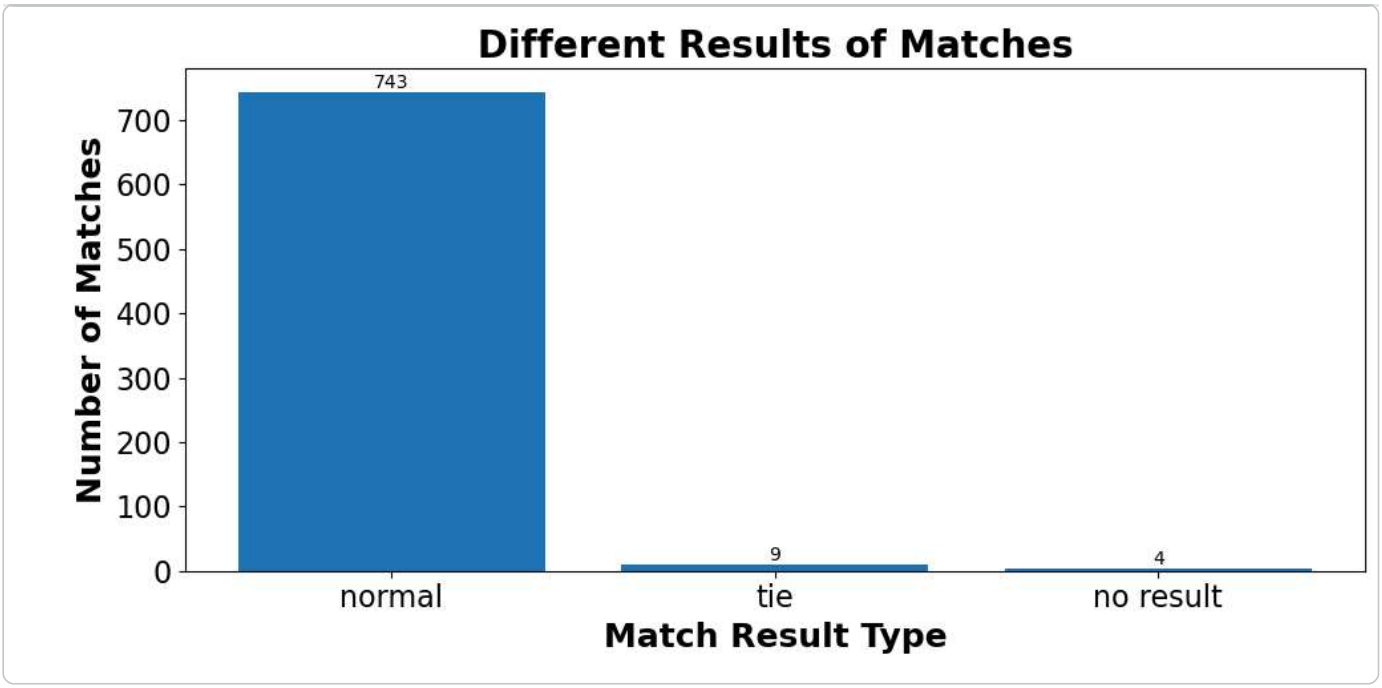


Insights

- This graph shows most of the seasons are affected due to rain.
- The most affected season due to rain was 2016 where results of four matches are affected by DLS method.
- According to this graph rain doesn't affect some seasons like 2010, 2012, 2013.
- This graph shows weather had limited impact on overall tournament.

13.Different results for matches.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("matches.csv")

result_counts = df['result'].value_counts()

plt.figure(figsize=(10,5))
plt.bar(result_counts.index, result_counts.values)

plt.xlabel("Match Result Type",fontweight='bold',fontsize=18)
plt.ylabel("Number of Matches",fontweight='bold',fontsize=18)
plt.title("Different Results of Matches",fontweight='bold',fontsize=20)

for i, value in enumerate(result_counts.values):
    plt.text(i, value, str(value), ha='center', va='bottom')

plt.xticks(rotation=0,fontsize=16)
plt.yticks(rotation=0,fontsize=16)
plt.tight_layout()
plt.show()
```

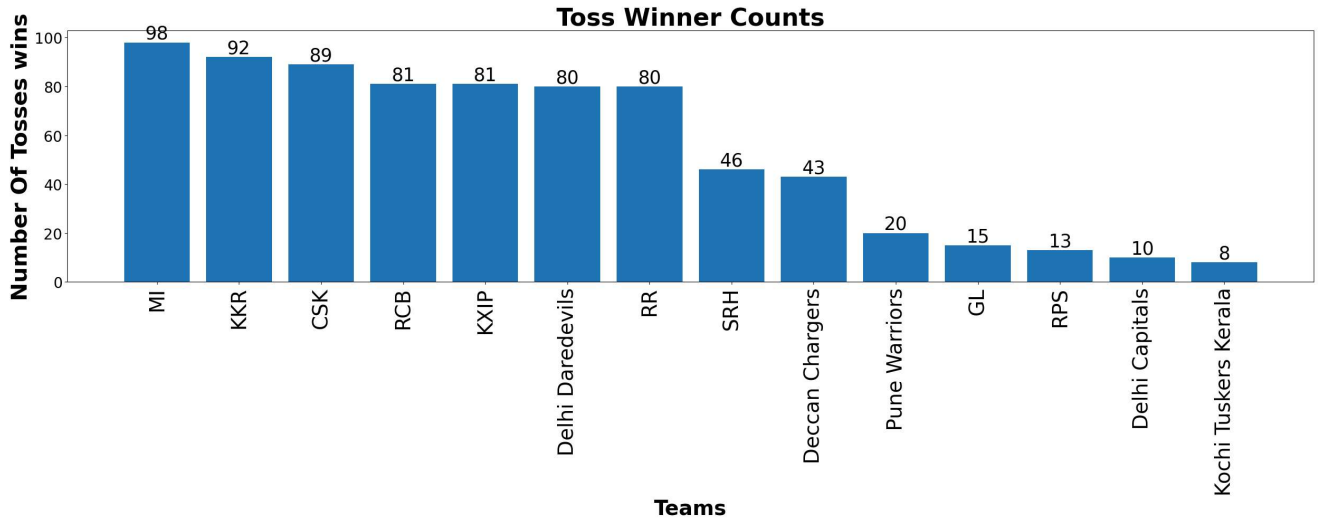


Insights

- Graph indicates that most of the resluts of matches are norman. Means winner or loser is decided.
- There are 9 matches which are tie means both teams scored equal.
- There are are 4 matches who has no result means they are intrrupted due to rain or other reasons.
- Most normal result shows effective scheduling, good pitch conditions.

14.How many times did a team win the toss?.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("matches.csv")

toss_winner_counts = df['toss_winner'].value_counts()

plt.figure(figsize=(25,10))
plt.bar(toss_winner_counts.index, toss_winner_counts.values)

plt.xlabel("Teams",fontsize=28,fontweight='bold')
plt.ylabel("Number Of Tosses wins",fontsize=30,fontweight='bold')
plt.title("Toss Winner Counts",fontsize=33,fontweight='bold')

for i, value in enumerate(toss_winner_counts.values):
    plt.text(i, value, str(value), ha='center', va='bottom',fontsize=25)

plt.xticks(rotation=90,fontsize=28)
plt.yticks(fontsize=20)
plt.tight_layout()
plt.show()
```



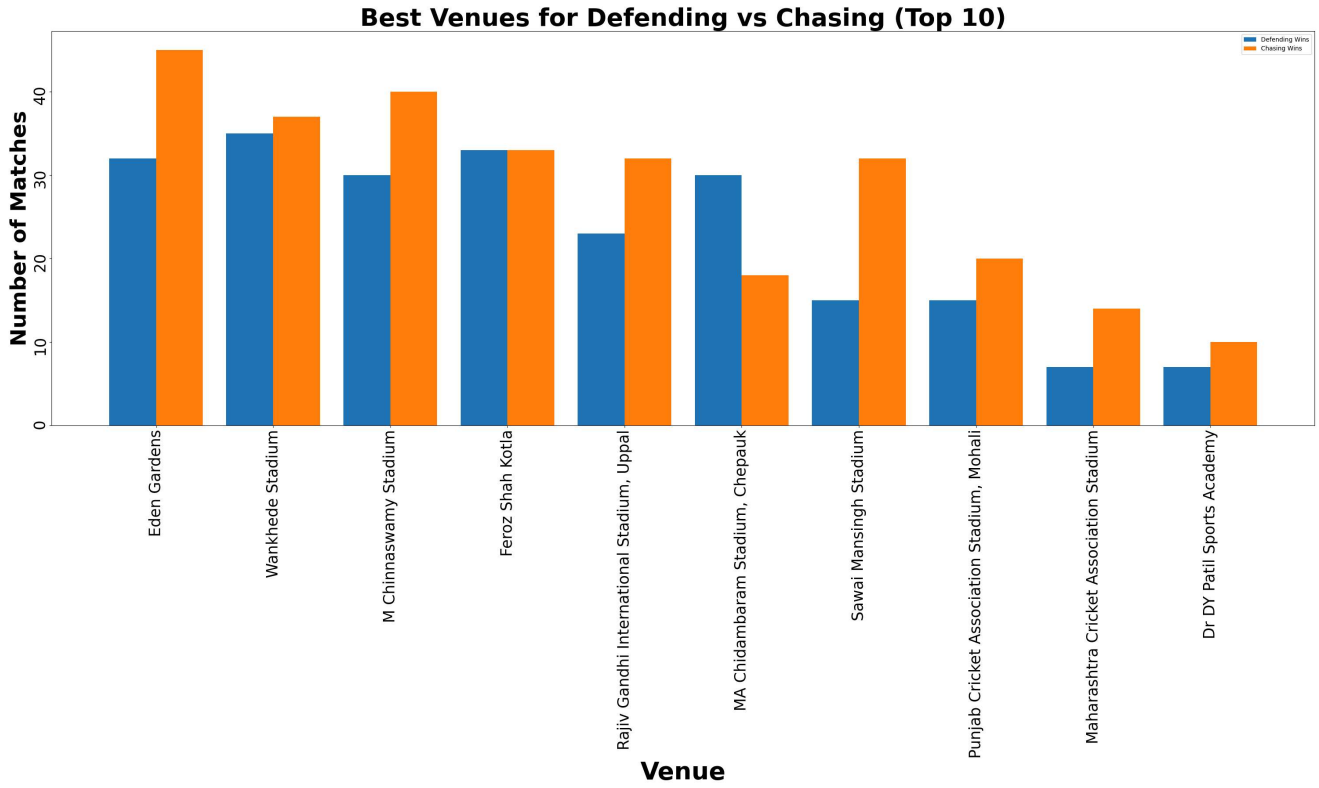
Insights

- Mumbai indians has won most (98) tosses means they hare more opportunity to choose what they want batting or bowling.
- KKR and CSK are behind MI which shows their consistent performance in tournament.
- RCB and KXIP along with DD and RR have won equal number of tosses shows they have equal opportunities.
- Last 5-6 teams has won very few tosses which shows their least participation in tournament.
- This graph shows first 7 teams have dominate the tournament.

15.Best venue for defending and chasing a total.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("matches.csv")

defend = df[df['win_by_runs'] > 0]
defend_counts = defend['venue'].value_counts()

chase = df[df['win_by_wickets'] > 0]
chase_counts = chase['venue'].value_counts()

venue_comparison = pd.DataFrame({
    'Defending Wins': defend_counts,
    'Chasing Wins': chase_counts
}).fillna(0)

venue_comparison['Total'] = venue_comparison.sum(axis=1)
venue_comparison = venue_comparison.sort_values('Total', ascending=False).head(10)

plt.figure(figsize=(30,18))

x = range(len(venue_comparison))
plt.bar(x, venue_comparison['Defending Wins'], width=0.4, label='Defending Wins')
plt.bar([i + 0.4 for i in x], venue_comparison['Chasing Wins'], width=0.4, label='Chasing Wins')

plt.xticks([i + 0.2 for i in x], venue_comparison.index, rotation=45)
plt.xlabel("Venue",fontsize = 40,fontweight='bold')
plt.ylabel("Number of Matches",fontsize = 35,fontweight='bold')
plt.title("Best Venues for Defending vs Chasing (Top 10)",fontsize = 40,fontweight='bold')
plt.legend()

plt.xticks(rotation=90,fontsize = 25)
```

```
plt.yticks(rotation=90,fontsize = 25)
plt.tight_layout()
plt.show()
```



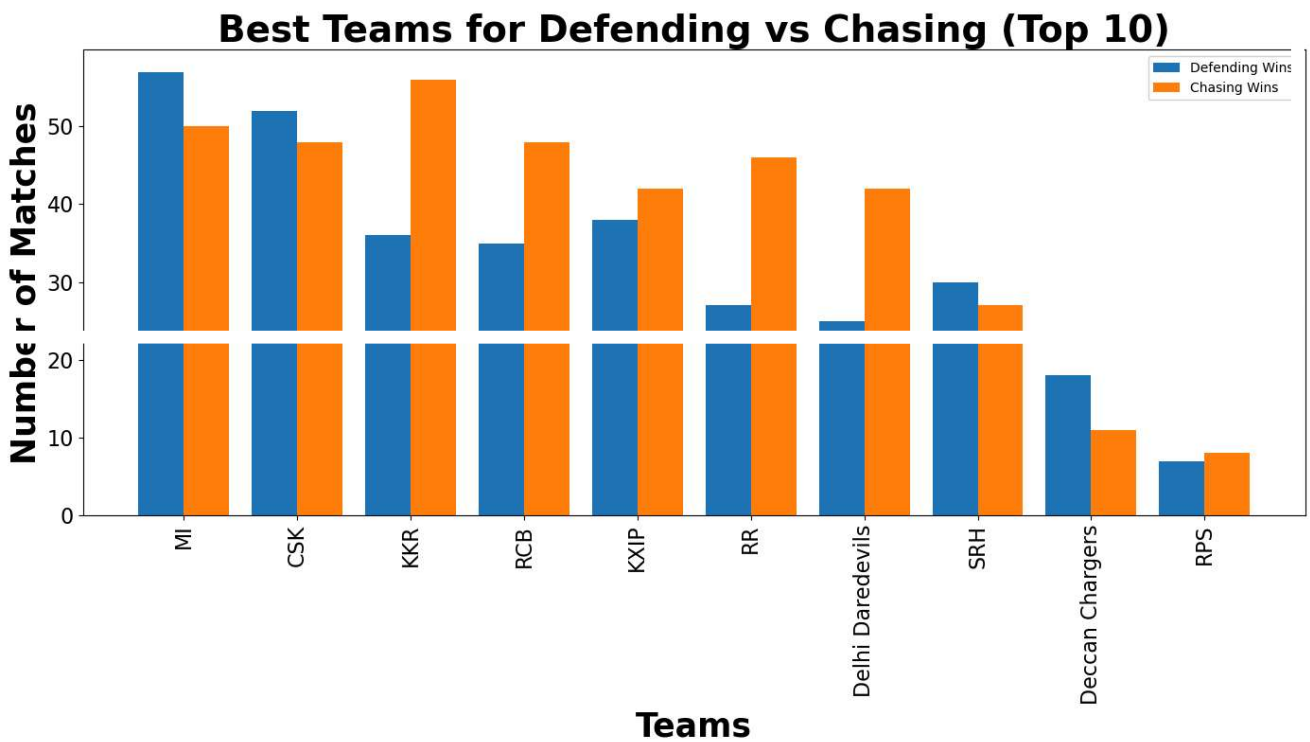
Insights

- Eden Gardens is easy for chasing teams, chasing wins (45) are much higher than defending wins (32), indicating teams prefer batting second here.
- Feroz Shah Kotla shows equal results for both defending and chasing wins are equal (33 each), making it a neutral venue.
- Only MA chidambaram stadium have higher defending wins than chasing wins means in these top 10 venues only this stadium favors defending wins.
- Sawai mansingh stadium have large difference in defending and chasing wins shows this stadium highly favors batting second.

16.Best defending and chasing teams.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("matches.csv")

defend = df[df['win_by_runs'] > 0]
defend_counts = defend['winner'].value_counts()

chase = df[df['win_by_wickets'] > 0]
chase_counts = chase['winner'].value_counts()

winner_comparison = pd.DataFrame({
    'Defending Wins': defend_counts,
```

```
    'Chasing Wins': chase_counts
}).fillna(0)

winner_comparison['Total'] = winner_comparison.sum(axis=1)

winner_comparison = winner_comparison.sort_values('Total', ascending=False).head(10)

plt.figure(figsize=(14,8))

x = range(len(winner_comparison))

plt.bar(x, winner_comparison['Defending Wins'], width=0.4, label='Defending Wins')
plt.bar([i + 0.4 for i in x], winner_comparison['Chasing Wins'], width=0.4, label='Chasing Wins')

plt.xticks([i + 0.2 for i in x], winner_comparison.index, rotation=45)
plt.xlabel("Teams",fontsize = 25,fontweight='bold')
plt.ylabel("Number of Matches",fontsize = 25,fontweight='bold')
plt.title("Best Teams for Defending vs Chasing (Top 10)",fontsize = 28,fontweight='bold')
plt.legend()

plt.xticks(rotation=90,fontsize = 16)
plt.yticks(rotation=0,fontsize = 16)
plt.tight_layout()
```



Insights